

Static games

1. Game 1

- Let $<$ denote strict dominance of a strategy. We assume common knowledge of rationality.
- $B < D$. Then, $d < a$ (Since B is not played, d is never best response to remaining)
- Then $AC < D$. Then $c < a$. Then $A < D$. Then $a < b$
- So only b, D remaining since IESDS arrives at a singleton set of undominated strategies of R_i for $i \in \{1, 2\}$ the game is dominance solvable.

~~For this part~~

~~for this part~~
~~analogous?~~

[Notes mention Pearce's Lemma, do we just state this?]

An action (pure strategy) is rationalizable if it's a best response to some mix of the opponent's rationalisable strategies. Since for each player there's one single strictly dominant strategy b, D ; that's the only rationalisable one.

Actually, probably wrong to say that, vs "IESDS-surviving"?

A NE is a strategy profile s.t. no agent has a strict unilateral incentive to deviate. Here, the unique NE is (D, b) , since they're IESDS survivors ~~strictly dominant~~ ~~other strategies are never best-responses~~.

Yes.

- Rationalizability is equivalent to surviving IESDS for a finite games, by Pearce's Lemma. When a game only has a single rationalisable strategy for each player (dominance-solvable) that is the unique (pure) NE.

Game 2

- Note that R is a never best response for column. It's strictly dominated by, e.g., a 50:50 mix of L and C . So eliminate it.
- Likewise, $B < (T:0.5; M:0.5)$.
- What remains is a zero-sum game (well, constant-sum). There's thus

✓ no pure-strategy NE. (We have $T, M; L, C$ Surviving IESDS, each are rationalizable strategies.)

✓ The single mixed NE is that each player ~~dominates~~ mixes 50:50 between their two rationalisable strategies, by indifference principle.

Yes... and anything else?

2i. The game is symmetric, so we can analyse from one player's perspective. Let the target value $T := \frac{p}{N} \sum x_i$.

• Suppose I choose ~~$x_i = 100$~~ $x_i = 100$. The "target" value $T = \frac{1}{Np} \sum x_i$ is either $= 0$ or > 0 , depending on whether everyone else chose 0 or not.

• If $T = 0$, then I'll be better-off choosing ~~100~~ regardless of others' choices.

✓ Note that $T \leq p \cdot \max \{x_i\}$ by construction. So I would be at least as close to the target by choosing 100p; my initial strategy is weakly dominated. Yes. This is important.

• Now suppose I choose $x_i = 100p$. But as above, nobody ~~prefers~~ ^{strictly} prefers $x_j > 100p$, so by choosing now $T \leq 100p^2$. So I'm at least as well-off choosing $100p^2$.

✓ Continuing iteratively, we'll be left with $x_i^* = p^* \cdot 100 \rightarrow 0$, as IESDS survivor.

ii. Let $S_{-i} = \sum_{j \neq i} x_j$. The BR is to choose x_i s.t.

~~$x_i = \frac{p}{N} S_{-i}$~~
$$x_i = \frac{p}{N} (S_{-i} + x_i) \text{ i.e. the exactly the target value.}$$

• So $x_i^* = \frac{p}{N-p} S_{-i}$

iii. (A) If all players choose $x_i = 0$, this is a NE: $T = 0$, so all players receive ~~$10/N$~~ $10/N$. There's no strict incentive to deviate since choosing any $x_i > 0$ means you're no longer (joint) closest to T and hence you receive 0. (You bring up the average, but not by more than $1/2$ of your ~~change~~ ^{change}, since other players stay at 0)

Err... Not sure about this.
You also need to rule out cases in which some choose 0 and others don't.

(b) There aren't any other NE. Suppose some players do not choose zero. (By symmetry, let's assume it's all of them) But then I can't be playing a best response to the others' actions, since $x_i^* < \bar{x}_{-i} = \frac{1}{N-1} S_{-i}$, where $0 < p < 1$ and $x_i^* = \frac{p}{N-p} S_{-i}$

Alternatively, note that $x_i^* = \left(\frac{p}{N-p} \cdot \sum_{j \neq i} x_j^* \right)$ by BR with each x_j^* identical and $= x_i^*$ where $\frac{p}{N-p} \neq 0$
 $\Rightarrow x_i^* = 0$

since $(N-p)x_i^* = (N-1)p x_i^*$
 $x_i^* = p x_i^*$

This needs justification...

3. Game 1 ✓ (B, R) is the only pure NE, by best-responses. $(1, 1)$ payoff
There's a mixed NE $\left(\frac{1}{2} T + \frac{1}{2} M, \frac{1}{2} C + \frac{1}{2} R \right)$. $(1, 1)$ E payoff

[Do you need to somehow show you found all NE? How?]

Consider a mix over all strategies: let col play L, C w.p. r, s ; row play T, M w.p. p, q . By indifference principle, we require

$2 \cdot s = 2 \cdot r = 1 \cdot (1 - s - r) \Rightarrow s = r$
 $2r = 1 - 2r \Rightarrow r = \frac{1}{4}$

and $2 \cdot p = 2 \cdot q = 1 \cdot (1 - p - q) \Rightarrow p = q = \frac{1}{4}$

So mixed NE of $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}; \frac{1}{4}, \frac{1}{4}, \frac{1}{2} \right)$, E payoff $\left(\frac{1}{2}, \frac{1}{2} \right)$

Technically: check mixes over every possible subset of actions for each player.
In practice, there will often be game-dependent shortcuts.

Game 2 This is a non-generic game, since pay col is indiff. between L, C in response to T.

Three pure NE: (T, L) , (T, C) , (B, R) .

Infinite mixed NE: $(T, pL + (1-p)C)$ for $p \in (0, 1)$

(All above have guaranteed $(1, 1)$ payoffs).
What about if row mixes? $\rightarrow$

Let row play T w.p. p ; col play L, C w.p. $q, 1-q$

At NE, $r + s = 1 - r - s \Rightarrow r + s = 1/2$

and $p = p = 1 - p \Rightarrow p = 1/2$

Nice

So $(\frac{1}{2} T + \frac{1}{2} B, r L + (\frac{1}{2} - r) C + \frac{1}{2} R)$ for $r \in (0, \frac{1}{2})$
NE with π payoffs $(\frac{1}{2}, \frac{1}{2})$.

Game 3

Two pure NE: $(T, L, 1)$ and $(B, R, 2)$ with payoffs
 $(1, 2, 3)$ $(3, 2, 1)$

MSNE: let each play T ; L ; 1 w.p. p, q, r .

P1: $r \cdot q = 3 \cdot (1-r) \cdot (1-q) \Rightarrow rqr = 3 - 3qr - 3r$

P2: $2 \cdot (p \cdot \frac{1}{2}) = 2 \cdot (1-p)(1-\frac{1}{2}) \Rightarrow pr = 1 - p - r + pr$

P3: $3 \cdot rqr = 1 \cdot (1-p)(1-q) \Rightarrow 2pq = 3 - 3p - 3q$

P2: $p = 1 - r$

$p = 1 - r$ by P2.

So $2(1-r)q = 3(1-(1-r)-q)$

$2 - 2rq = 3 - 3 + 3r - 3q$

$5q - 2qr = 3r$

$5q - 3r = 3 - 3q - 3r \Rightarrow q = \frac{3}{8}$

Sub into P3: $2pq = 3 - 3p - 3q$

Yeah. Reasoning is correct - just double-check algebra.

$p = 3 - 3p - q/q \Rightarrow q = 1/2, r = 1/2$

So NE is $(\frac{1}{2} T, \frac{3}{8} L, \frac{1}{2} B)$ with π payoffs $(\frac{1}{2}, \frac{1}{2})$

Must be missed up algebra.

...+ give an argument to show that there are no other NE.

(strictly)

- 4.1. T is dominant if regardless of Col's strategy, the payoff for Row is always (strictly) greater when they play T vs B.

This holds when $a > c$ and $b > d$. ✓

ii. Matching pennies:

1, -1	-1, 1
-1, 1	1, -1

[Correct to use strict ineq?]
Suppose neither player has a dominant strategy. That implies:
• $(a < c \text{ and } b > d)$ or $(a > c \text{ and } b < d)$, AND
• $(\alpha < \gamma \text{ and } \beta > \delta)$ or $(\alpha > \gamma \text{ and } \beta < \delta)$ ✓
Must be either red or green ✓

Yes, otherwise you get a pure NE.

This would be sufficient for no PSNE. If it's the case that only one player has a dominant strategy, then the other has a (weak) best response available in a pure strategy* - so there would be a PSNE (but perhaps also MSNE). So the conditions above are necessary.
* Since they can determine what opponent optimally plays

iii. Indifference principle: let them play T; L u.p. p, q .

$$P1: q \cdot a + (1-q)b = q \cdot c + (1-q)d$$

$$P2: p \cdot \alpha + (1-p)\beta = p \cdot \gamma + (1-p)\delta$$

✓ So $q = \frac{d-b}{a-b-c+d}, p = \frac{\delta-\beta}{\alpha-\beta-\gamma+\delta}$

iv. We have an MSNE provided $p, q \in (0,1)$. Suppose there's no PSNE. Consider p, q :

• By ii, numerators are nonzero.

• $(a+d) - (b+c)$ will be -ve in case 1, where num is also -ve.

In case 2, it's +ve and num is +ve. (Same for q .)

• Num always smaller in abs. val. than denom, e.g. 1/2

• so they're valid probabilities; MSNE exists given no PSNE.

or in case 2 $a-c > 0$
so $(d-b) < (d-b) + (a-c)$, etc.